

L6: The Notion of "Node Importance"

Now that we have defined a number of centrality metrics, the obvious question is: *which metric should we use for each network analysis application?*

To choose the right metric, it is important to understand the notion of “**node importance**” that each of these centrality metrics focuses on.

- **Degree (or strength) centrality:** (<https://gatech.instructure.com/courses/552990/pages/l6-degree-eigenvector-and-the-katz-centrality>) these metrics are more appropriate when we are interested in the number or weight of direct connections of each node. For example, suppose that you try to find the person with most friends in a social network.
- **Eigenvector/Katz/PageRank centrality:** (<https://gatech.instructure.com/courses/552990/pages/l6-degree-eigenvector-and-the-katz-centrality>) these metrics are more appropriate when we are mostly interested in the number of connections with other well-connected nodes. For example, suppose that you analyze a citation network and you try to identify research papers that are not just cited many times – but they are cited many times by other well-cited papers. For undirected networks, it is better to use Eigenvector centrality because it does not depend on any parameters. For directed networks, use Katz or PageRank depending on whether it makes sense to split the centrality of a node among its outgoing connections. For instance, this splitting may make sense in the case of Web pages but it may not make sense in the case of citation networks.
- **Closeness (or harmonic) centrality:** (<https://gatech.instructure.com/courses/552990/pages/l6-closeness-centrality-and-harmonic-centrality>) these metrics are more appropriate when we are interested in how fast a node can reach every other node. For example, in an epidemic network, the person with the highest closeness centrality is expected to cause a larger outbreak than a person with very low closeness centrality. If the network includes multiple connected components, it is better to use harmonic centrality.
- **Betweenness centrality metrics:** (<https://gatech.instructure.com/courses/552990/pages/l6-betweenness-centrality-variants>) these metrics are more appropriate in problems that involve some form of transfer through a network, and the importance of a node relates to whether that node is in the route of such transfers. For example, consider a data communication network in which you want to identify the most central router. You should use a betweenness centrality that captures correctly the type of routes used in that network. For instance, if the network uses shortest-path routes, it makes sense to use shortest-path betweenness centrality. If however, the network uses equally all possible routes between each pair of nodes, the path centrality may be a more appropriate metric.